Multimodal NGS analysis of gene expression and chromatin state in the olfactory epithelium

UBDS^3 | 2026

Tutor: Lisa Korneeva
TA: Olexandr Lykhenko

Your project team

Lisa Korneeva - Project lead
PhD fellow at Boulard Group, EMBL Rome

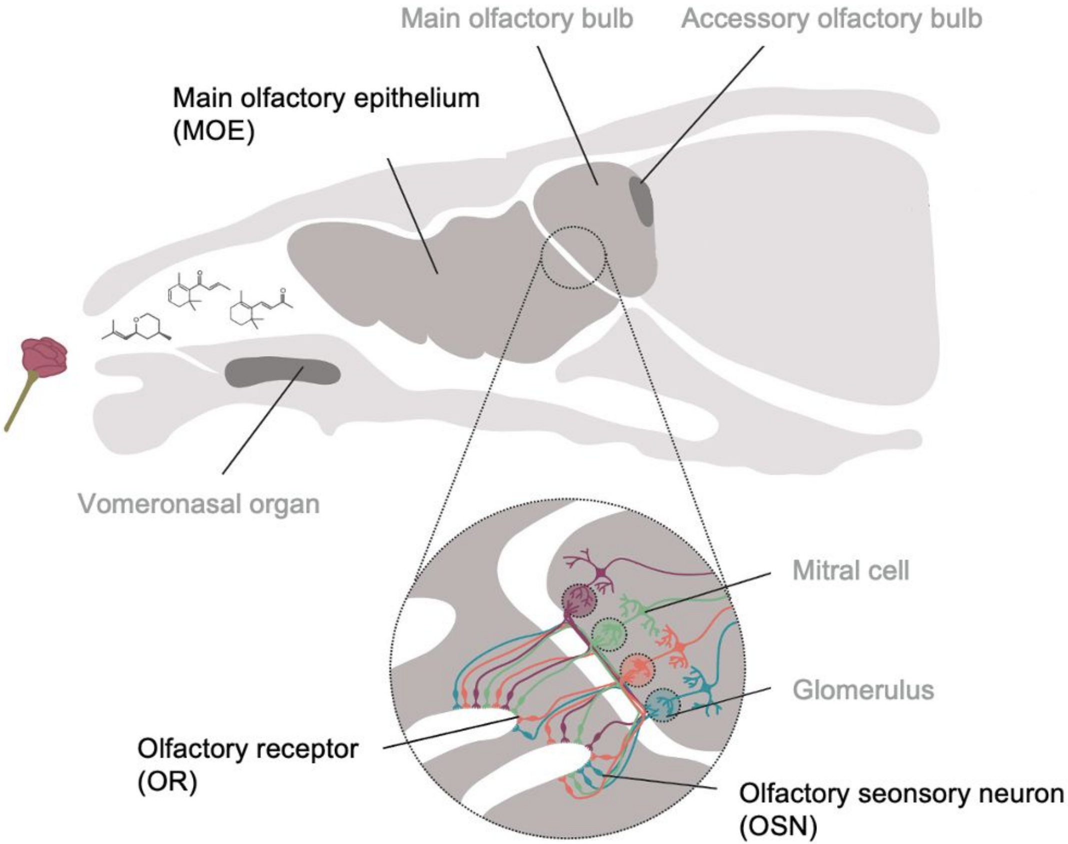
Olexandr Lykhenko - Teaching assistant
Junior researcher at Systems biology group, Institute of Molecular Biology and Genetics of NASU, Kyiv, Ukraine

Project goal

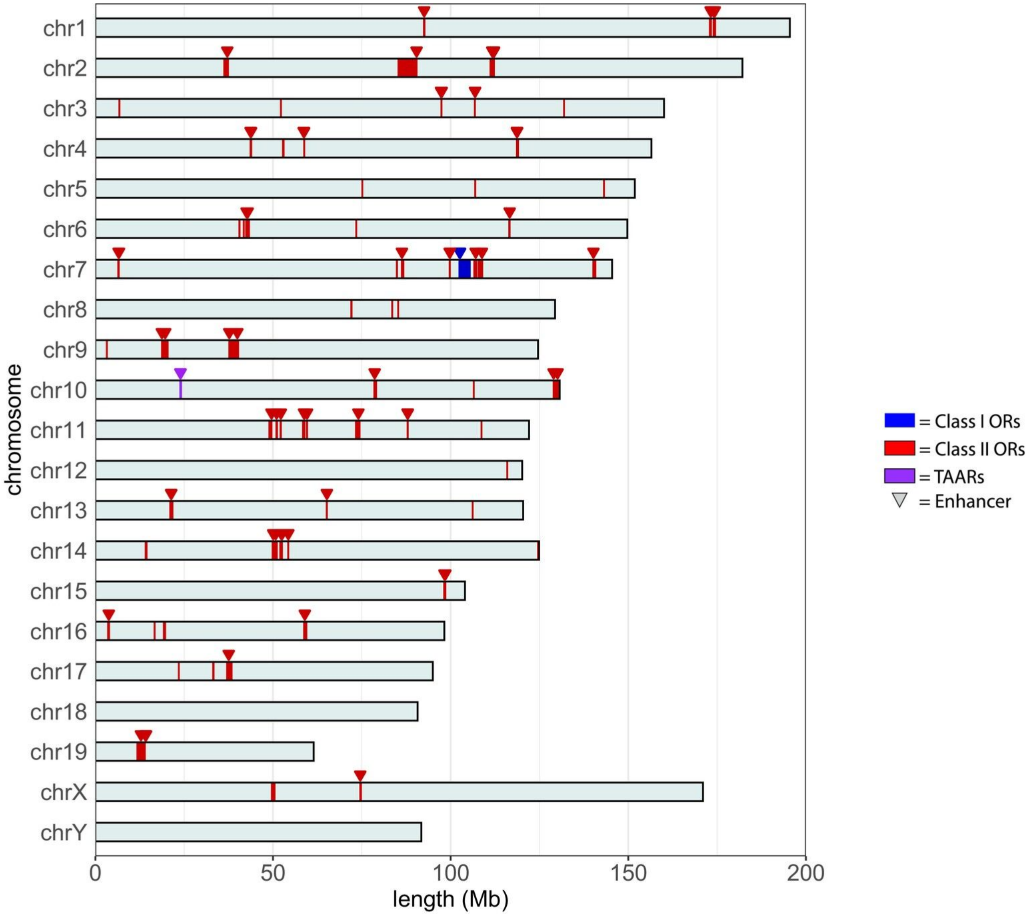
Follow the full journey of sequencing data from biological sample to biological insight

Olfactory system

- Around 1,300 olfactory receptor genes in mice (around 400* in humans)
- Understudied mechanisms of regulation
- My PhD project, so I am biased

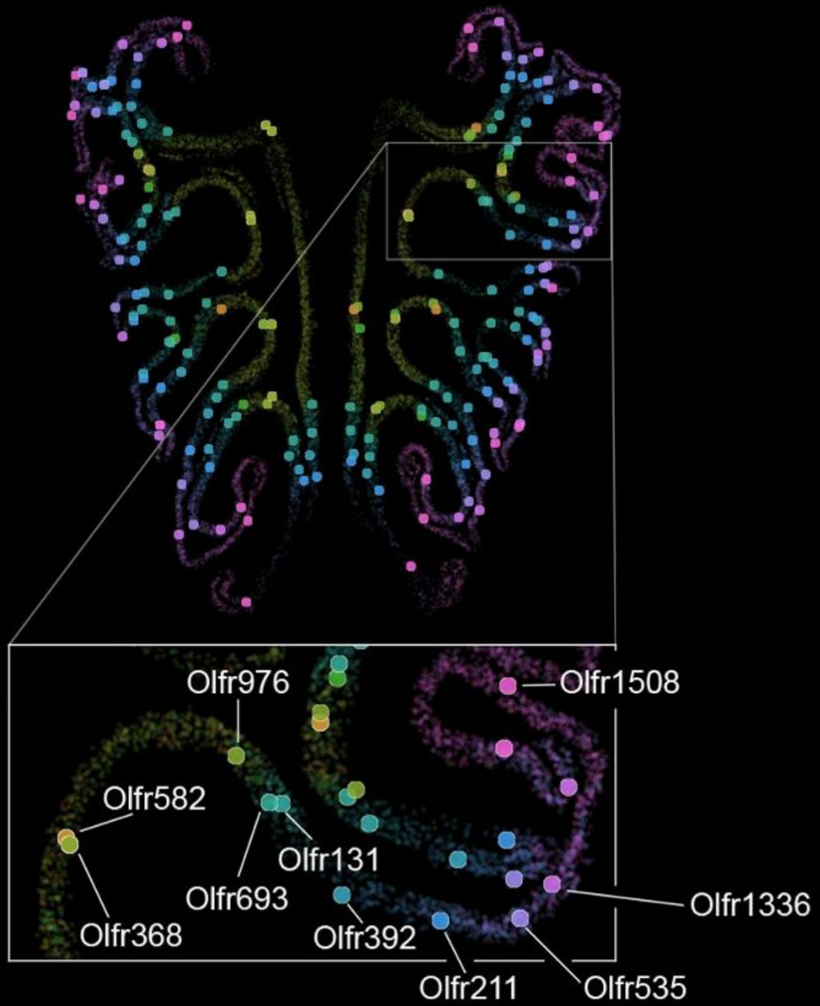


(Adapted from Bear et al., 2016)



Olfactory receptor genes are the largest gene family in mammals

(Yusuf & Monahan, 2024)



https://www.the-scientist.com/a-smell-map-reveals-insights-about-olfactory-organization-74447

Nobel Prize in Physiology / Medicine 2004

Richard Axel and Linda B. Buck

Why 1 receptor per neuron?

Why spatial patterning?

Why epigenetic signatures?

How can we approach the system with diverse regulatory mechanisms on the transcriptomic and epigenetic levels?

Next Generation Sequencing (NGS)

A high-throughput technology that “reads” the nucleotide sequence of DNA or RNA fragments. You can profile an entire genome, transcriptome, or epigenome, making it the backbone of modern genomics research.

RNA-seq

A method for measuring gene expression genome-wide by sequencing RNA molecules (normally converted into cDNA) from a cell or tissue.

ChIP-seq

A method for mapping the genomic locations of specific proteins or histone modifications by sequencing DNA that co-precipitates with an antibody target.

Course structure

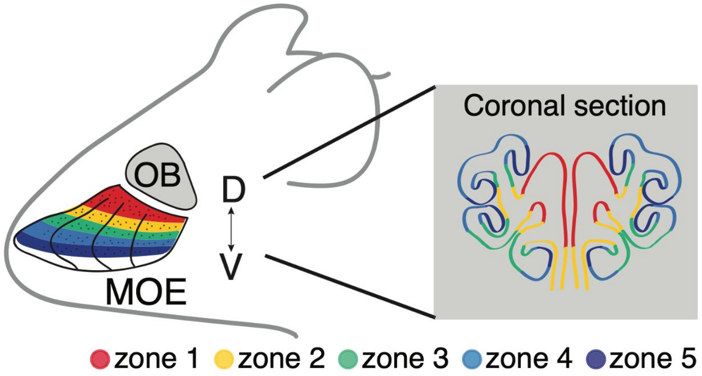
1. Publicly available RNA-seq and ChIP-seq dataset

2. 3 pipelines:
   - Downloading the data
   - RNA-seq
   - ChIP-seq (hopefully)

3. Lecture materials on next-generation sequencing technology and steps that take place before the data analysis

4. Guided tutorials on pipeline execution and analysis roadmap

Publicly available RNA-seq and ChIP-seq dataset



eLife
RESEARCH ARTICLE

Opposing, spatially-determined epigenetic forces impose restrictions on stochastic olfactory receptor choice

Elizaveta V Bashkirova[1,2], Nell Klimpert[3], Kevin Monahan[4], Christine E Campbell[5,6], Jason Osinski[5,6], Longzhi Tan[7], Ira Schieren[2], Ariel Pourmorady[1,2], Beka Stecky[2], Gilad Barnea[3], Xiaoliang Sunney Xie[8,9], Ishmail Abdus-Saboor[2], Benjamin M Shykind[10], Bianca J Marlin[2], Richard M Gronostajski[5,6], Alexander Fleischmann[3], Stavros Lomvardas[2,11]*

Pipelines: Snakemake / Nextflow

- **fetchngs** to download the data: https://nf-co.re/fetchngs/1.12.0/
- **3t-seq** for RNA–seq data analysis: https://boulardlab.github.io/3t-seq/
- **nf-core ChIP-seq** for ChIP-seq data analysis: https://nf-co.re/chipseq/2.1.0

By the end of the course you will:

- Understand the main principles behind **NGS technologies** and library preparation
- Be able to set up and run **RNA-seq** and **ChIP-seq** analysis pipelines
- Know how to assess **data quality** and interpret QC reports
- Perform **differential gene expression analysis** and visualise results in **R**
- Understand how chromatin state and gene expression relate to each other
- Have hands-on experience with reproducible computational workflows using **Snakemake** and **Nextflow**

Prerequisites

- Number of people: **ideally up to 5** (we can see what we can do if more people want to join).
- Bash command line: **familiar with** navigating the filesystem, running scripts, and basic syntax from the command line.
- R: **intermediate** - comfortable with data manipulation (**dplyr, tidyverse**), basic plotting (**ggplot2**), and reading/writing files. Experience with **DESeq2** is a plus.

Looking forward to exploring NGS data analysis with you!

:)